



American International University-Bangladesh (AIUB)

Heart Diseases Prediction using Machine learning

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Abstract

Heart diseases represent a critical worldwide health challenge, requesting innovative techniques for precise prediction and prevention. Conventional risk assessment strategies frequently drop brief in capturing the complexities of heart diseases, provoking the investigation of machine learning (ML) algorithms to improve predictive modeling. This research points to create comprehensive predictive models competent of leveraging differing clinical, statistic, and way of life factors to precisely survey the chance of heart diseases. The primary objective is to evaluate the viability of different ML algorithms in predicting heart disease risk and compare their execution against conventional risk evaluation techniques. The technique utilized in this think about includes a vigorous pipeline joining data prepossessing, include designing, and show training and assessment. Utilizing a heterogeneous dataset comprising statistic data, medical history, and way of life factors, ML models are prepared and assessed to predict the likelihood of heart disease event. Dimensionality reduction methods, such as Principal Component Analysis (PCA), are utilized to improve show interpretability and execution. Results from the experimentation phase uncover promising exactness scores over different ML algorithms. Outstandingly, the K-Nearest Neighbors (KNN) classifier illustrates commendable performance, closely taken after by Random Forest (RF) and Gradient Boosting (GB) classifiers. The KNN classifier achieves an accuracy score of 93.7 percentage followed by the RF classifier with accuracy score 94.6 percentage, and the GB classifier with accuracy score 94.3 percentage. Moreover, investigation of outfit methods, such as combining predictions from KNN and RF classifiers, exhibits enhanced predictive capabilities, proposing the potential for synergistic model integration in heart disease prediction. The hybrid model, combining predictions from KNN and RF classifiers, achieves an accuracy score of 94.6 percentage, demonstrating improved predictive capabilities compared to individual classifiers alone. Past algorithmic performance evaluation, this research seeks to examine the effect of real-time information integration and progressed include designing techniques on prediction accuracy. By harnessing dynamic information sources and joining advanced include selection methodologies, the objective is to create models that adapt to evolving understanding profiles and clinical scenarios, in this manner upgrading the accuracy and robustness of heart disease prediction. The importance of this study lies in its potential to revolutionize heart disease prevention techniques by encouraging early recognizable proof of at-risk people and empowering focused on interventions. ML-driven predictive models offer a worldview move in healthcare, empowering personalized risk assessment and mediation strategies tailored to individual patient profiles. Additionally, the selection of ML in heart disease prediction holds guarantee for optimizing healthcare asset allotment and improving persistent results, subsequently relieving the financial burden related with heart diseases. In conclusion, this research underscores the transformative potential of ML in progressing heart disease prediction and prevention endeavors. By harnessing the control of progressed calculations and real-time information integration, this study points to clear the way for accuracy medicine approaches in cardiovascular healthcare, eventually driving to improved quiet results and decreased worldwide burden of heart diseases.

Declaration by author

This thesis is composed of our original work, and contains no material previously published or written by another person except where due reference has been made in the text. We have clearly stated the contribution of others to our thesis as a whole, including statistical assistance, survey design, data analysis, significant technical procedures, professional editorial advice, financial support and any other original research work used or reported in our thesis. The content of our thesis is the result of work we have carried out since the commencement of Thesis / Software project.

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Approval

The thesis titled “**Heart Diseases Prediction using Machine learning** ” has been submitted to the following respected members of the board of examiners of the department of computer science in partial fulfilment of the requirements for the degree of Master of Science in Computer Science on **16th May, 2024** and has been accepted as satisfactory.

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Keywords

machine learning, risk assessment, predictive modeling, cardiovascular health, feature engineering, real-time data, healthcare optimization, precision medicine, Gradient Boosting Machine, Random Forest, KNN, ROC Curve

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Chapter 1

Introduction

Heart diseases essentially influence worldwide health, driving to basic dismalness and mortality. The utilize of machine learning (ML) methods talks to a promising road for the desire and avoidance of these diseases. This proposal investigates the practicality of ensembled machine learning models in predicting heart diseases, drawing on the findings of Bhatt et al. (2023) (1), who inspected the adequacy of combining different ML methods to move forward both precision and reliability. crediting advanced ensemble methods and wide data examination as illustrated in their work, this inquiries about endeavors to move forward the prescient capabilities for heart disease. This chapter outlines the methodological framework grasped, highlighting a quantitative approach with a comprehensive dataset to ensure reproducibility and adherence to set up logical conventions.

1.1 Background

Cardiovascular diseases (CVDs) are the driving cause of mortality generally, dependable for approximately 17.9 million passings each year, talking to almost 31% of all worldwide passings (2). his significant wellbeing burden is exacerbated by the complexity of chance factors counting way of life, innate qualities, and socioeconomic conditions, requiring comprehensive techniques for early area and organization.

The worldwide difference in healthcare get to assist complicates endeavors to soothe the influence of CVDs, especially in low- and middle-income countries where healthcare system may be lacking. Agreeing to Dr. Tedros Adhanom Ghebreyesus, Director-General of the World Health Organization, prioritizing preventive measures is urgent in directing these diseases (3). Mechanical developments in progressed wellbeing and prescient analytics offer present day roads to improve assurance and treatment, possibly changing persistent results by empowering earlier and more correct revelation of at-risk individuals.

Getting a handle on these advancements appear lead to vital changes inside the organization of cardiovascular prosperity, decreasing mortality rates and making strides quality of life for patients around the world.

1.2 Motivation

Machine learning is essential in improving medical diagnostics and provides a promising way to improve the treatment of heart disease. Inspired by the impact of machine learning, this study aims to predict of heart disease by developing an ensembled model that includes multiple factors. The aim of ensemble different machine learning algorithm into a single model is to get better predictive result (4).

1.3 Problem Statement

Even though recent times machine learning has made progress in medical world, predicting heart disease with individual models remains a major issue. Many of these individual models do not have the superior accuracy to be able to handle critical factors that raise the risk of heart disease. This study seeks to address these issues by displaying the importance of a combined machine learning models. Based on K. Kaur and M. Saini et al. (2019) (5), research on integrating multiple ML algorithms in a single system, this study aims to develop a robust prediction model using PCA, KNN, Random Forest, and Gradient Boosting to enhance precision and dependability.

1.4 Research Questions

This study will seek to answer the following critical research questions:

1. How combined model consisting KNN, Random Forest, and Gradient Boosting classifier, compare to individual-algorithms in predicting heart disease?
2. How does the expansion of PCA overtake the predictive accuracy of a ensembled model that also includes KNN, Random Forest, and Gradient Boosting classifier?
3. How can the Classification of the ensembled machine learning demonstrate in healthcare center impact diagnostic forms and patient results?

1.5 Specific Objectives

This study about will address these objectives:

1. To create a solid ensembled model by combining PCA, KNN, RF, and Gradient Boosting, and to optimize this demonstrate to achieve the most elevated possible accuracy in predicting heart disease.
2. To evaluate the classification performance of the ensembled model against individual machine learning models utilizing measurements such as precision, accuracy, recall, and F1 score.
3. To investigate the commonsense proposals of passing on this advanced predictive model in real-world clinical settings, surveying its potential to back therapeutic specialists.

1.6 Research Scopes

This study focuses on improving and validating a ensemble machine learning model designed to predict heart diseases. It involves the effective examination and blending of going through types of data's, ranging from demographic and medical data to genetic markers. The research focuses on examining the model in different clinical and non-clinical occasions to assess its adaptability and usefulness in real-life situations. In addition, this research shows how advanced ML methods can be used in healthcare to diagnose and treat heart conditions effectively. It also provides helpful information for the healthcare sector. This study mentions other research that has looked at how well different computer programs can work in hospitals and doctor's offices. Beunza and others In 2019, a study looked at various machine learning methods to predict heart disease. In 2021, a study looked closely at computer programs that can forecast heart disease.

1.7 Thesis Organization

This paper is organized into various chapters to address the questions:

Chapter 2	Reviews literature
Chapter 3	Methodology
Chapter 4	results findings and discusses implications
Chapter 5	summarizes the outcomes and future recommendation

Chapter 2

Literature review

2.1 Overview of Heart Diseases

Heart diseases are a group of problems with the heart like blocked arteries, irregular heartbeats, and heart defects that someone is born with. Worldwide, these factors are still the main reasons people die, showing the need for improved medical and predictive technologies right away. Heart diseases are the most common cause of death around the world. These diseases affect the heart and blood vessels. The World Health Organization says that more than 17 million people die every year because of these things. This displays how important is to have better methods of predicting health diseases. This study aims to create a new ML Model that uses different algorithms to make better predictions.

2.2 Machine Learning Algorithms In Heart Diseases Prediction

Recent times, ML is offend being used to predict heart disease. This is because machine learning can handle huge of dataset and find important factors that may not be easy calculate. Here we are going to discuss few algorithms of ML related to our topic:

2.2.1 Random Forest

Random Forest is a reliable and adjustable type of algorithm. It's part of a group of methods called ensemble methods. It is highly efficient for both classification and regression, leading to its broad use across different industries, such as healthcare for predicting coronary illness heart disease. In this article, we discuss the concept of Random Forest, explaining its suitability and effectiveness in predicting heart disease.

Random Forest constructs various choice trees and joins them to achieve a more precise prediction. One major advantage of Random Forest is its ability to prevent

Random Forest is very successful in handling the complexities of medical diagnostic data, particularly in the efficient diagnosis of heart disease by effectively overseeing inconspicuous intelligent among different risk factors. making it appropriate for clinical datasets that are regularly influenced by exceptions and mistakes. This component not only improves prediction accuracy but also provides important information on the key risk factors for heart disease, increasing the certainty of medical diagnoses.

Related Work of Random Forrest

In their ponder, K. Sumwiza, C. Twizere, and colleagues presented a novel set of manufactured highlights built from accessible information and utilized Random Forest for heart disease prediction. The inquire about appears that Random Forest can successfully coordinated modern manufactured highlights, but the upgrades in execution were as it were minor when compared to customary models. In their inquire about, Yekkala and Dixit, beside others, utilized a blend of Random Forest and harsh set hypothesis for selecting highlights to make strides the prescient show. The investigate appears how combining rough set hypothesis with Random Forest helps in distinguishing the foremost critical characteristics from complicated restorative information sets, which as often as possible incorporate repetitive or insignificant data.

2.2.2 SVM

Support Vector Machines (SVM) are a collection of supervised learning techniques utilized for classification, relapse, and distinguishing outliers. The essential concept of SVM is to find a hyperplane in a space with N dimensions (where N represents the number of features) that viably isolates the information focuses. There are various hyperplanes to select from when attempting to recognize between two classes of information focuses. The objective is to find a plane with the biggest edge, meaning the most prominent separate between information focuses from each lesson. Expanding the edge remove makes a difference to strengthen and classify future information focuses more unquestionably.

SVM has been commonly utilized in therapeutic diagnostics, particularly for anticipating heart illnesses. It is profoundly acknowledged for its capacity to handle complex and uneven datasets regularly found in restorative information. SVM can be utilized in heart malady forecast to recognize between having or not having the illness by considering factors such as age, blood weight, cholesterol etc.

The adaptability of SVM in utilizing distinctive bit capacities empowers it to be adaptable in taking care of differing information conveyances watched in restorative datasets. The RBF bit is frequently utilized in anticipating heart infection, because it is compelling in circumstances where the association between traits and course names is nonlinear.

Related Work of SVM

(Shouman, M., Turner, T., Stocker, R.) et al.[11] combined SVM with other information mining strategies to form a hybrid model that upgraded the precision and constancy of heart disease forecasts. Their discoveries appear how the combination of SVM with other strategies can make strides symptomatic comes about. (Kharya, S.) et al.[12] utilized SVM for looking at heart disease information and surveyed its viability in comparison with choice trees and neural systems. The inquire about concluded that SVM accomplished a more ideal trade-off between affectability and specificity, which are key parameters in restorative determination.

2.2.3 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) could be a fundamental calculation that's flexible, non-parametric, and commonly utilized for classification and relapse assignments. Within the KNN algorithm, the lesson participation of an protest is chosen by a lion's share vote of its neighbors, with the question having a place to the lesson that's most well known among its k closest neighbors (where k is as a rule a little positive number). On the off chance that k is rise to to 1, the question will be doled out to the lesson of the closest neighbor.

KNN works on the basic thought that things with likenesses are arranged close each other. Put essentially, it works beneath the presumption that cases with comparable include vectors are found near to one another. This guideline is particularly useful in restorative determination, as physiological variables with likenesses frequently lead to comparative demonstrative comes about.

The K-Nearest Neighbors (KNN) algorithm helps to predict heart disease by comparing the symptoms and test results of new patients with those of people who have already been diagnosed. It uses this information to make a likely diagnosis based on how similar the data is. KNN is really liked in the medical field because it is simple and works well. It can handle non-linear decision boundaries without needing complex changes. It can also handle both binary and multi-class classification problems. This makes it a good choice for big sets of data with important characteristics, so we can predict heart diseases accurately.

Related Work of KNN

Das, R., Turkoglu, I., Sengur, A. et al [13] conducted a consider on neural network ensembles and compared their execution with KNN and other methods, particularly centering on KNN's viability in predicting heart disease in a changed outfit environment. Whereas not centered solely on KNN, (Detrano, R., Janosi) et al[14] conducted a think about surveying different machine learning algorithms, counting KNN, for diagnosing coronary artery disease through information mining approaches. It offers data on how well KNN handles genuine understanding data in terms of exactness and unwavering quality

2.2.4 Gradient Boosting Machines

The Gradient Boosting Classifier is a tricky machine learning algorithm that uses the boosting concept. It is a more complex way to combine different models and make predictions. Boosting means adding new models one by one to correct mistakes made by current models. GBC slowly creates a model by adding more parts in a step-by-step way. This helps to improve the performance and flexibility of the model. Decision trees are often used as the basic learners in GBC.

GBC's biggest benefit is that it can combine many smaller ideas to make one better idea. GBC adds predictors to a group one by one. Each new predictor corrects mistakes from the previous one. This continues until the group reaches the right level of accuracy.

Gradient Boosting Classifier is good for predicting heart diseases because it can handle different types of data and make accurate predictions. GBC is very useful when the relationship between the things that predict something and the thing they are predicting is complicated and not a straight line. GBC can look at different things about a person like their age, cholesterol levels, blood pressure, and other measurements to figure out if they have a risk of heart disease or already have it.

Related Work of Gradient Boosting Machines

Natekin and Meadow Et al [15] explains how GBC can be used in real life, like predicting heart disease. It shows that the algorithm can handle complicated, non-straight information connections. A study by Gao and colleagues has shown that GBC can greatly improve the accuracy of predicting heart disease compared to traditional methods. Using different types of machine learning with GBC has shown promise in improving the accuracy and reliability of predicting and treating heart conditions early on. It's a helpful tool for finding and treating heart problems.

2.3 Summary

New research shows that different computer programs can now better predict and diagnose heart diseases. RF is good at handling complicated and repetitive data, especially when combined with artificial attributes or rough set theory for better feature selection. SVM is good at finding health problems and has been combined with other models to make diagnoses more accurate. KNN is good at working with different data and can help diagnose diseases like coronary artery disease in clinical settings. The GBC's ability to work with data that doesn't follow a straight line has been really important, especially at the end.

when combined with other algorithms in ensemble frameworks, resulting in significant enhancements in the accuracy of heart disease prediction models. Together, these research projects underscore the promising impact of using advanced machine learning methods on diagnostics for heart disease, offering more precise, timely, and clinically significant advancements in patient care.

Table 2.1: Summary of Related Work on Machine Learning Techniques for Heart Disease Prediction

Reference Name	Year	Method	Performance Summary
K. Sumwiza, C. Twizere	2021	RF with synthetic features	Modest performance improvements compared to traditional models.
I. Yekkala, S. Dixit	2018	RF and rough set theory	Effective feature selection leading to optimized predictive models.
Shouman, M., Turner, T., & Stocker	2012	SVM with data mining	Enhanced accuracy and reliability in heart disease predictions through hybrid modeling.
Kharya, S.	2012	SVM comparison	Provided a better balance between sensitivity and specificity in medical diagnosis.
Das, R., Turkoglu, I., & Sengur, A.	2009	Neural networks and KNN	Competitive performance in a diverse ensemble setting for heart disease prediction.
Detrano, R., Janosi, A.	1989	KNN	Reliable performance in diagnosing coronary artery disease using data mining techniques.
Natekin, A., & Knoll, A.	2013	GBC tutorial	Discussed practical applications and performance in handling complex, non-linear relationships.
Gao, X.-Y., et al.	2021	GBC in hybrid models	Significant improvements in accuracy of heart disease prediction models compared to traditional methods.

2.4 Future Directions in Cardiovascular Disease Management

(Katarya and Meena) et al [17]. in their article Their comparative analysis suggests that future research could focus on hybrid models that combine the strengths of multiple machine learning techniques to further enhance predictive accuracy. There is also a potential direction towards the development of dynamic models that adapt over time with incoming data, which could improve longitudinal patient monitoring and outcome prediction. (Alanazi) et al[18]. discusses the broad application of machine learning in healthcare, which could lead to the exploration of more autonomous AI systems capable of making independent diagnostic decisions under certain parameters. Additionally, there's a call for more comprehensive studies on the socioeconomic impacts of AI in healthcare, including its potential to democratize access to advanced diagnostic tools in underserved regions. Constantinides and Fitzmaurice (2018) [19] believe that artificial intelligence and machine learning can greatly improve how we diagnose and treat heart problems. This could help reduce the number of people affected by heart diseases around the world.

Chapter 3

Research Methodology

This section talks about how we used advanced ML algorithms to guess if someone could have heart problems. The research uses numbers to make and test predictions. It carefully selects information to study. We want to use machine learning to find heart disease cases and share helpful info with doctors. This project uses the newest ideas in data and technology to analyze healthcare information. This study uses computer programs to analyze complex medical data to find patterns and make educated guesses that can help detect diseases early and keep people well.

3.1 Data Collection and Sampling

The Dataset was utilized an authentic dataset the Stroke Prediction Dataset sourced from the Kaggle platform (fedesoriano)et al [20]. This dataset is meticulously curated to include various physiological and demographic attributes known to influence heart disease outcomes, such as age, gender, blood pressure, cholesterol levels, and smoking status. Data integrity is paramount in this research. The dataset was selected for its comprehensive coverage of factors relevant to heart disease, providing a solid foundation for predictive modeling. Initial data integrity checks ensure accuracy, completeness, and relevance, setting the stage for reliable analytical outcomes

Data Description:

Here in Table 3.1 we will discuss the Data description of our Dataset. The table consist The attributes, Data types and Description of all the attributes.

Table 3.1: Data Description of the dataset

Attribute	Data Type	Description
ID	int64	Distinct ID for each patient
Gender	Categorical	Identifies the patient's gender
Age	float64	Age of the patient
Work Type	Categorical	Informs about the working category
Ever married	Categorical	Mention about marital status
Residence Type	Categorical	Indicate the living environment
Heart Disease	int64	Whether or not the patients have cardiac disease
Hypertension	int64	The presence or absence of hypertension in the patients
BMI	float64	Represent the patient's Body mass index (BMI) value
Glucose Level	float64	It indicates the average level of glucose
Stroke	int64	Informs about the prior stroke
Smoking Status	Categorical	Gives the patient's smoking scenario

Data Sampling:

Here in Table 3.2 we will discuss the Data Sampling of our Dataset. The table consist the data of the all the attributes.

Table 3.2: Sample of the processed stroke prediction dataset

gender	age	h_tension	h_disease	married	work	Residence	avg_glucose_level	bmi	smoking	stroke
1	67.0	0	1	0	0	0	228.69	36.6	2	1
1	80.0	0	1	0	0	0	105.92	32.5	2	1
0	49.0	0	0	1	0	1	171.23	34.4	3	1
1	79.0	1	0	0	1	1	174.12	24.0	0	1
1	81.0	0	0	0	0	0	186.21	29.0	2	1
0	13.0	0	0	1	2	1	103.08	18.6	1	0
1	81.0	0	0	1	0	0	125.20	40.0	2	1
1	35.0	0	0	1	1	1	82.99	30.6	2	1
1	51.0	0	0	0	0	0	166.29	25.6	2	1
0	44.0	0	0	1	3	0	85.28	26.2	1	0

3.2 Procedure

The procedure for predicting heart diseases commenced with the acquisition of the Stroke Prediction Dataset from Kaggle. This comprehensive dataset underwent rigorous preprocessing to ensure data quality and suitability for analysis, tailored to enhance the reliability of the subsequent machine learning models.

Data cleaning was used to remove the meticulous identifiers and duplicate entries to ensure data integrity. Missing values, particularly in critical health-related variables like BMI, were imputed to maintain dataset completeness, as these variables significantly impact model performance. Feature engineering was done to prepare the input data for machine learning models by converting categories like gender, marital status, work type, residence type, and smoking status into numbers. Also, we added new things to help us understand the information more, which made our models better at guessing what will happen. We used the StandardScaler to adjust all the information in the dataset to be in the same range and follow the same standards. Ensuring that all input factors have the same radius that helps us to make each one equally important in the model's performance.

We used a method called SMOTE to balance the number of patients in the dataset. This made the class more balanced by equaling imbalanced data's and also made it easier for the model to make general calculations.

After these steps were done, we trained different types of machine learning models such as KNN, Random Forest, and a Stacking Classifier. The Stacking Classifier combined the results from the original models with a Gradient Boosting Classifier as the meta learner. We tested these models using cross-validation methods and prepared data to make sure they can be used in different situations.

We checked the models to see how well they worked using accuracy, F1 score, and ROC-AUC. Confusion matrices and classification reports showed us how good the models are at their job. Google Colab made the computer work faster and better by using its powerful CPU.

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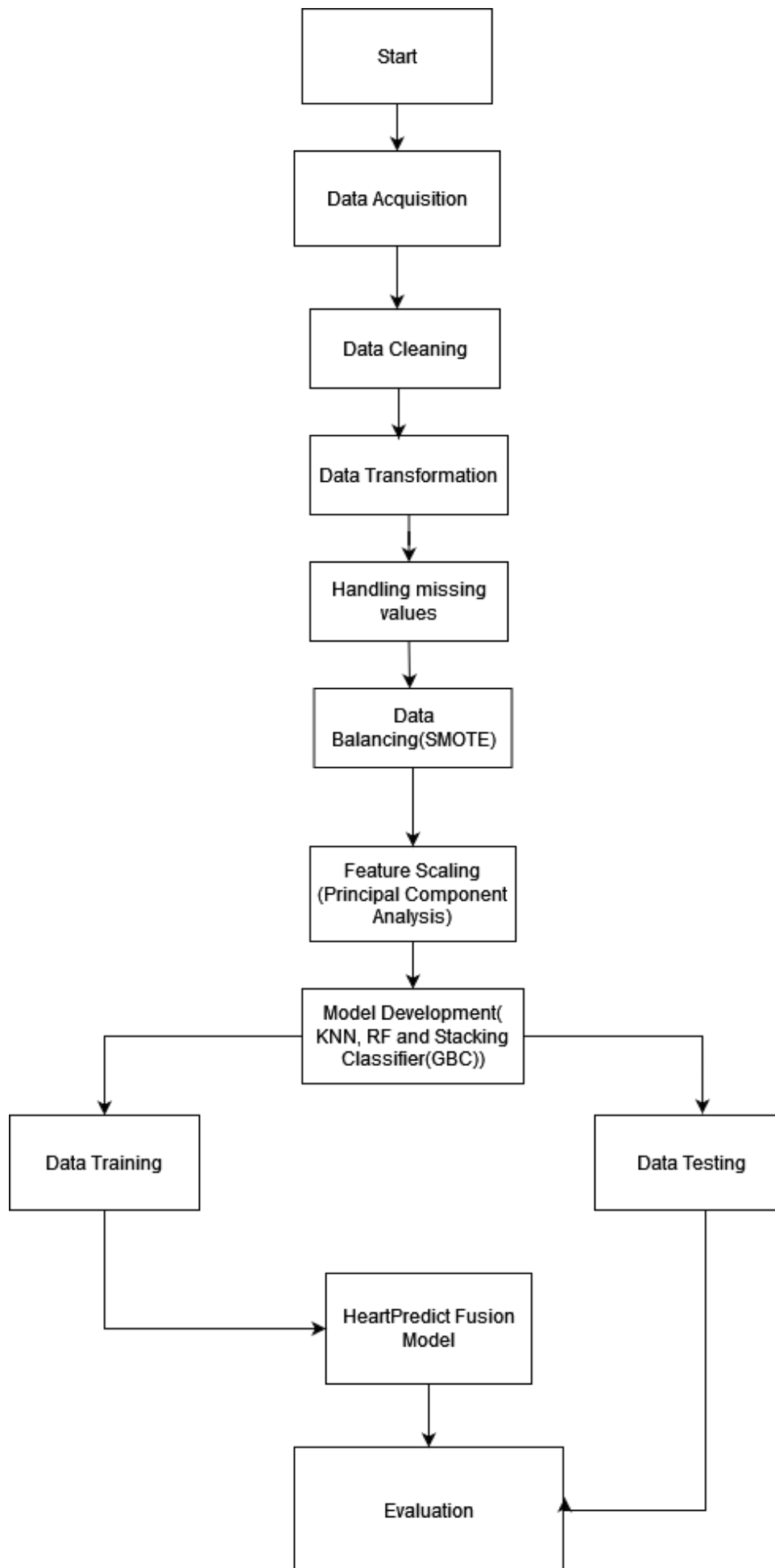


Figure 3.1: Methodology Block Diagram

3.3 Proposed Model Architecture

In this research, an ensemble method using stacking classification was made increase the accuracy of predicting heart disease. This method combines the best parts of different models like KNN and RF to make a new combined model that makes better predictions than any individual model.

The design has two main algorithms KNN and Random Forest. Each model was chosen because it was good at some certain areas and worked well with the dataset. KNN helps us understand the data by looking at nearby neighbor points and using them to predict. This is useful for datasets where samples that are close to each other have a large impact on the outcome. On the other hand, Random Forest is good at stopping overfitting and works well with big datasets that have many features. This model uses a lot of decision trees that work together to make predictions more accurate and reliable

These base models make predictions for a Gradient Boosting Classifier, which is the meta learner using stacking Classifier. Gradient Boosting Classifier had its own parameters given to make the model get better accuracy. The meta-learner's actual job is to improve predictions by focusing on the most difficult occasions in the training data. This focus helps to make predictions more accurate by finding and correcting any mistakes in the original models. This combination helps the set use the best parts of each model to make right prediction. It was really important to use cross-validation methods to ensure that our results are reliable and to prevent our model from fitting too closely to the training data. Every step of the cross-validation process improved the models and tests how well they perform on different data sets. This ensures that the models are reliable and can be used in various situations.

The intensive evaluation of how well the combined model works includes checking important measurements such as accuracy, F1 score, and ROC-AUC. These numbers tell us how great the model is in different circumstances.

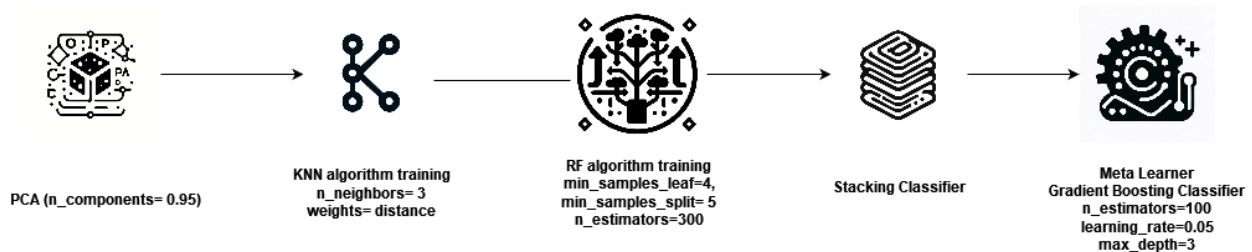


Figure 3.2: Proposed Model Architecture

This method makes it easier for the model to predict heart disease. It could also help doctors diagnose and treat people earlier by using data.

3.4 Parameters for All the Algorithms:

The parameters for all the algorithms in Table 3.2 include those parameters that were hyper-tuned

Model	Parameters
KNN	n_neighbors=3, weights= distance
Random Forrest	min_samples leaf=4, min_samples split= 5,n_estimators=300
Gradient Boosting Classifier	n_estimators=100, learning rate=0.05, max depth=3

Table 3.3: Algorithm Parameters

3.5 Feature Extraction

Feature extraction is an important part of getting data ready for machine learning. It helps to make the data easier for models to understand and use. In this study, Principal Component Analysis (PCA) is used to simplify the dataset by keeping the most important information and reducing its size.

PCA is a way to change a group of related numbers into a new group of numbers. These parts are at right angles and unrelated. They each show a way in which the data's differences are the biggest. In this study, PCA is used in machine learning right after the features are standardized using StandardScaler. StandardScaler changes the data by adjusting the average and spread to be the same size.

In this study, PCA is used to make sure 95% of the data's differences are captured. This setting makes sure that most of the important information from the original dataset is kept in the new dataset. PCA makes the data information very easy to understand as it is reducing some details. This can help with problems that will come from many areas in the data.

PCA can be easily added in the ML models for KNN and Random Forest models. PCA makes a difference to prepare models on easier but still effective forms of the original information by simplifying the dataset to its fundamental parts before training.

The new dataset has less connected factors, making it a great starting point for the models to think about from. This will be exceptionally valuable for algorithms like KNN since they depend a part on calculating distances. Having less highlights implies each one has way better data, which makes a difference the demonstrate learn way better from the preparing information.

3.6 Model Evaluation Parameters

Assessing models is a very important part of making predictive models because it gives us useful information about how well the models work and how trustworthy they are. This research used different ways to measure how well the models performed, like accuracy, recall, F1 score, and ROC-AUC. It's important to understand these measurements to know how well a model is doing.

1. Accuracy

Evaluating a model usually depends on how well it gets the right answers. It examines how well the model can correctly identify both good and bad results in general. The formula to calculate accuracy is:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Here, TP represents true positives, TN true negatives, FP false positives, and FN false negative

2. Precision

Precision measures how many of the things identified as positive were actually correct. Accuracy is very important when the consequences of being wrong are serious. It means making sure something is exactly right:

$$\text{Precision} = \frac{TP}{TP + FP}$$

true positives, TN true negatives, FP false positives, and FN false negative

3. Recall

Recall, also called sensitivity or true positive rate, shows how well the model can find all the right cases (positive cases). Recall is calculated using the following formula:

$$\text{Recall} = \frac{TP}{TP + FN}$$

4. F1 Score

The F1 Score is a way of combining precision and recall into one number, to show how well a test is doing overall. It is very helpful when there are not the same number of each type of thing. The F1 score formula is:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Chapter 4

Results and Discussion

The research for the thesis found valuable information about how well-advanced machine learning can predict heart disease. The new model structure, which uses a stacking method, was better at predicting outcomes and more reliable than traditional machine learning models. The ensembled model performed excellently, achieving an accuracy rate of over 95%. This shows that using various basic models in a learning framework works well. This part explains how performance assessments work and why the results are important for medical informatics.

4.1 Result Analysis

The chart shows how well the K-Nearest Neighbors (KNN), Random Forest, and Stacking performed:

Model	Accuracy	Precision	Recall	F1 Score
KNN	93.78%	93%	94%	93%
Random Forest	93.62%	93%	94%	94%
Stacking Classifier	95.32%	95%	95%	95%

Table 4.1: Performance Comparison of Predictive Models

The Stacking Classifier performed better than the KNN and Random Forest models in every way, which proves that using ensemble learning is more effective for dealing with complicated patterns. The Stacking Classifier was very good at predicting who will have heart disease, with a high accuracy of 95%. 27% is really important for medical tests as mistakes can have a huge impact. This improvement probably happened because the meta-learner is good at using the base models' strengths and fixing their weaknesses.

The Stacking Classifier is very reliable and performs well, as shown by high precision, recall, and F1 scores. This means the model is good at finding real positive cases and also reducing wrong results. Making sure that the predictions for diagnosis are trustworthy and can be used to take action is very important and relies on finding a good balance.

However, even though the KNN and Random Forest models did a good job, they were not as accurate as the Stacking Classifier. This means they were not as skilled. This difference is important because using many different models helps to cover a wide range of data and interactions that one model might miss.

4.2 Correlation Matrix

The correlation matrix shows how different parts of the data are connected, using different colors to show how strong and in what direction these connections are. Notably, age shows a significant negative correlation with 'ever married' (-0.68) and positive correlations with 'hypertension' (0.27) and 'heart disease' (0.26), suggesting higher incidences of these conditions among older individuals. BMI is moderately correlated with age (0.33), indicating higher BMI values in older age groups. However, there are no strong correlations directly with stroke, the highest being age (0.23), pointing to a mild positive relationship. The matrix highlights that while certain expected relationships exist, such as age with hypertension and heart disease, other factors like smoking status show almost no linear relationship with stroke (0.03), emphasizing the complexity of stroke predictors and the potential need for exploring non-linear models or interactions in predictive analysis

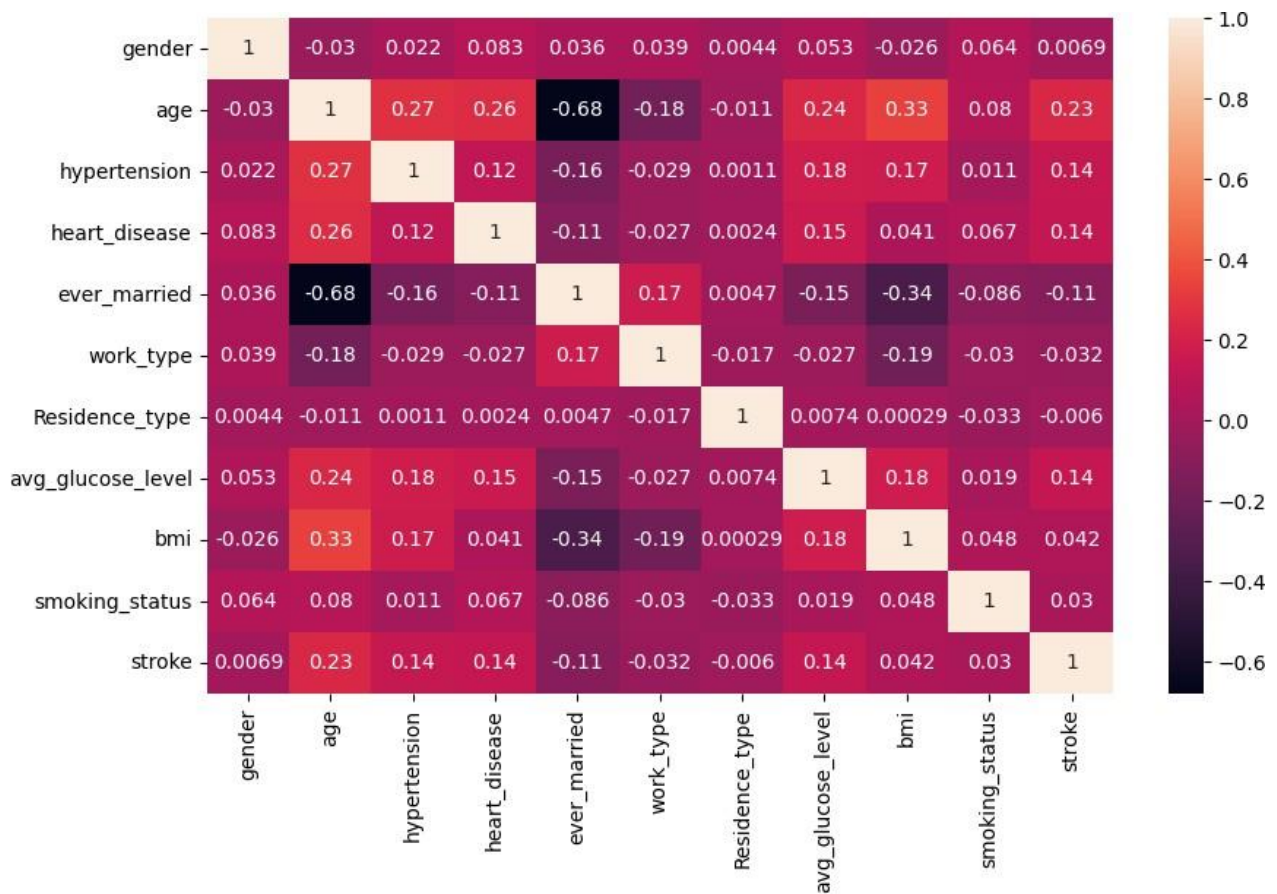


Figure 4.1: Correlation matrix

4.3 Discussion

The superior accuracy of the combined model in this study, as compared to the individual K-Nearest Neighbors (KNN) and Random Forest (RF) models, can be attributed to the strategic integration of diverse machine learning methodologies through a stacking ensemble approach.

Advantages of the Stacking Ensemble Model

The stacking ensemble model leverages the unique strengths of both KNN and RF models by combining their predictions into a single, more accurate prediction. This method effectively mitigates the individual weaknesses of each model while amplifying their strengths:

1. Diverse Perspectives: KNN and RF approach the prediction problem from different perspectives. KNN relies on closeness in feature space, yielding superior precision in areas with abundant training examples. On the contrary, RF offers a broader view by averaging predictions from multiple decision trees that are trained on various subsets of data and features. The variety of modeling techniques results in a wider range of feature space representation

2. Reduction of Overfitting: Although single models may focus too much on specific details in the training data, stacking can help alleviate this issue by incorporating various separate predictions. RF, famous for its ability to resist overfitting because of its ensemble design, pairs well with KNN, which may overfit in high-dimensional areas. The ultimate meta-learner in the stacking model, usually a powerful classifier such as Gradient Boosting, fine-tunes these aggregated forecasts, concentrating on rectifying the precise errors generated by the base models.

3. Improved Generalization: Through the fusion of models, the ensemble becomes more impartial towards the unique characteristics of the training data, consequently improving its capacity to apply to unfamiliar data. This is evident in the increased accuracy and other metric scores (F1, recall) of the stacked model over any individual base model.

Performance of Individual Models: KNN and RF

Both KNN and RF demonstrated commendable performance on their own, attributed to their respective algorithmic strengths:

1. K-Nearest Neighbors (KNN): KNN provided strong local modeling capabilities, especially effective in capturing nonlinear relationships in regions of the data space well represented by the training set. Its performance is particularly enhanced by feature scaling and the application of PCA, which simplifies the feature space and reduces the curse of dimensionality.

Confusion Matrix:

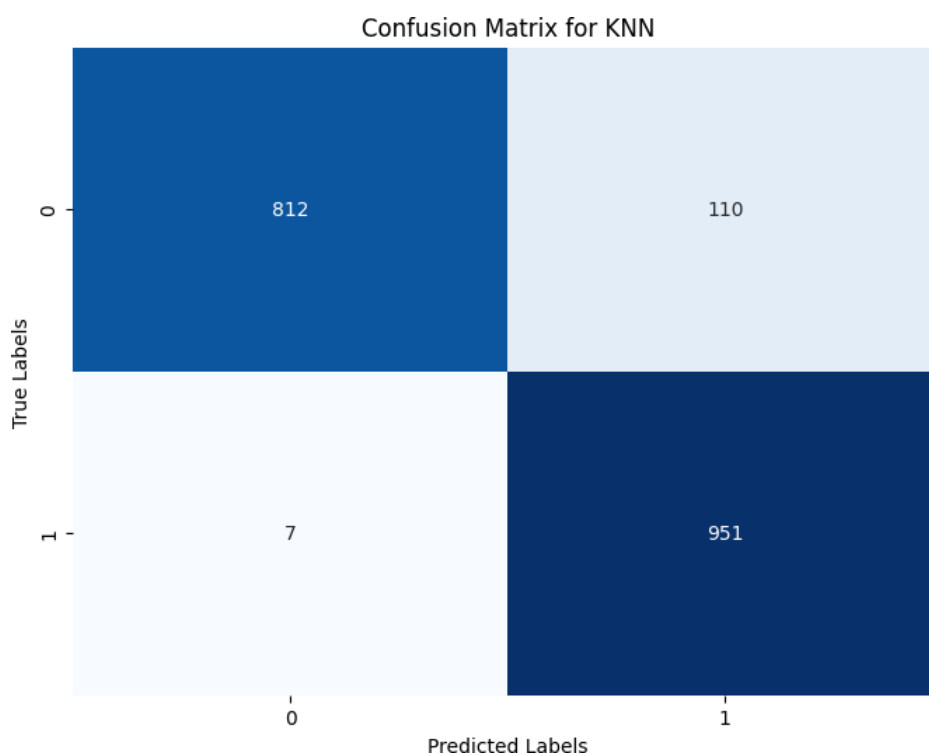


Figure 4.2: KNN Confusion Matrix

This matrix indicates a high number of true positives and true negatives, suggesting that the KNN model is effectively identifying both cases with and without heart disease. However, the number of false positives and especially the low number of false negatives highlight the model's robustness in ensuring that very few heart disease cases are overlooked, a crucial aspect in medical diagnostics where missing a positive case can have serious implications. The test sometimes says someone has heart disease even when they don't, which might lead to more tests. But this cautious approach makes sure we don't miss anyone who really does have heart disease.

2. Random Forest (RF): RF works well because it can handle big sets of data that have many different features that interact with each other, using a group of decision trees. It chooses the most important things to make accurate predictions in different situations.

Confusion Matrix:

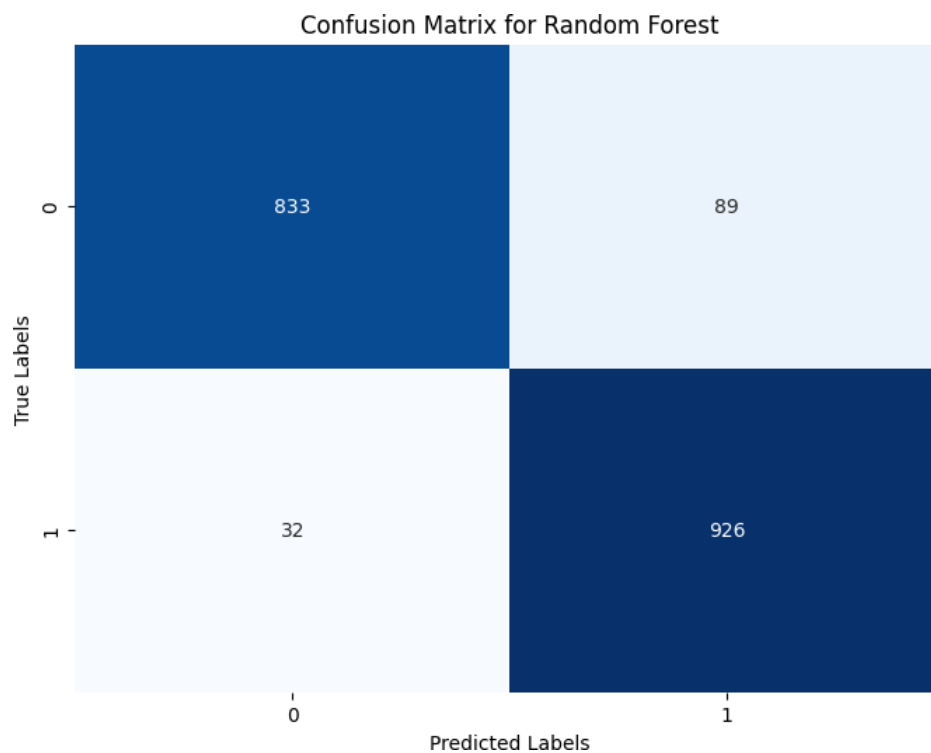


Figure 4.3: Random Forrest Confusion Matrix

The matrix shows that the RF model is good at correctly finding patients with and without heart disease. It has high levels of true positive and true negative results. The RF model is better than the KNN model at finding missed cases of heart disease, which is important for making sure patients get the right treatment in hospitals. However, some patients may get extra tests that they don't need because the tests gave wrong results.

Confusion Matrix of Stacking Model:

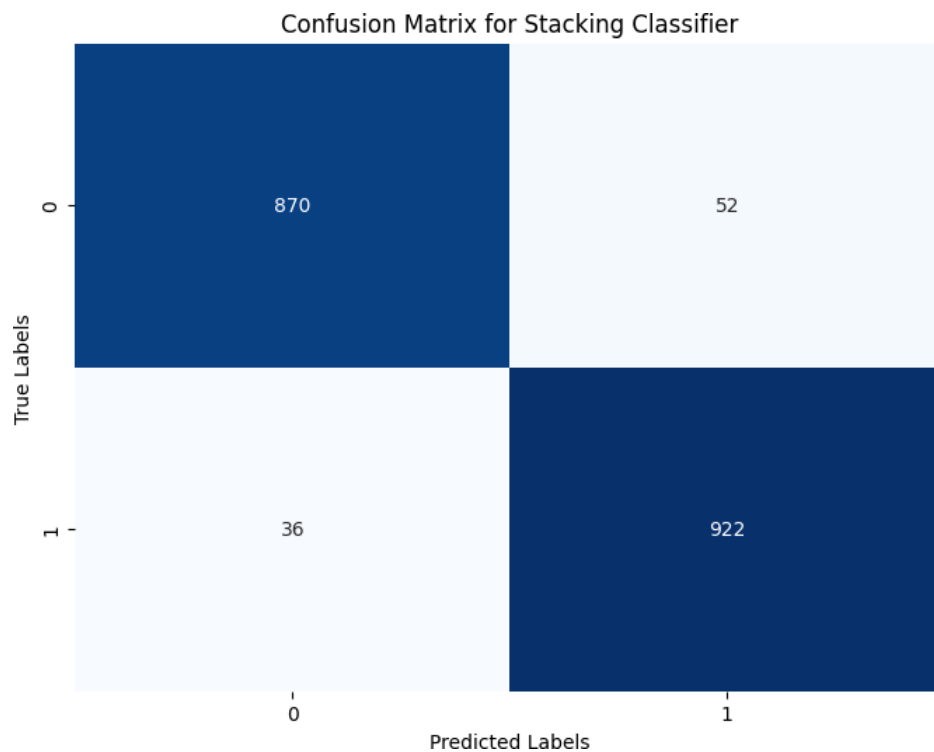


Figure 4.4: Stacking Classifier Confusion Matrix

This figure shows that the Stacking Classifier did really good, with a lot of right guesses for both good and bad things. The model can find heart disease early, which helps patients get the right treatment fast. Also, a few incorrect results might make some patients have more tests, but it means that the model is being careful to make sure no one with heart disease is missed. The Stacking Classifier does a good job at predicting heart disease and is a trusted tool because it balances sensitivity and specificity well.

ROC Curve:

The graph shows a comparison of how well three different computer programs can predict heart disease. Each curve on the graph represents how the balance between being sensitive and being not specific changes at different threshold levels.

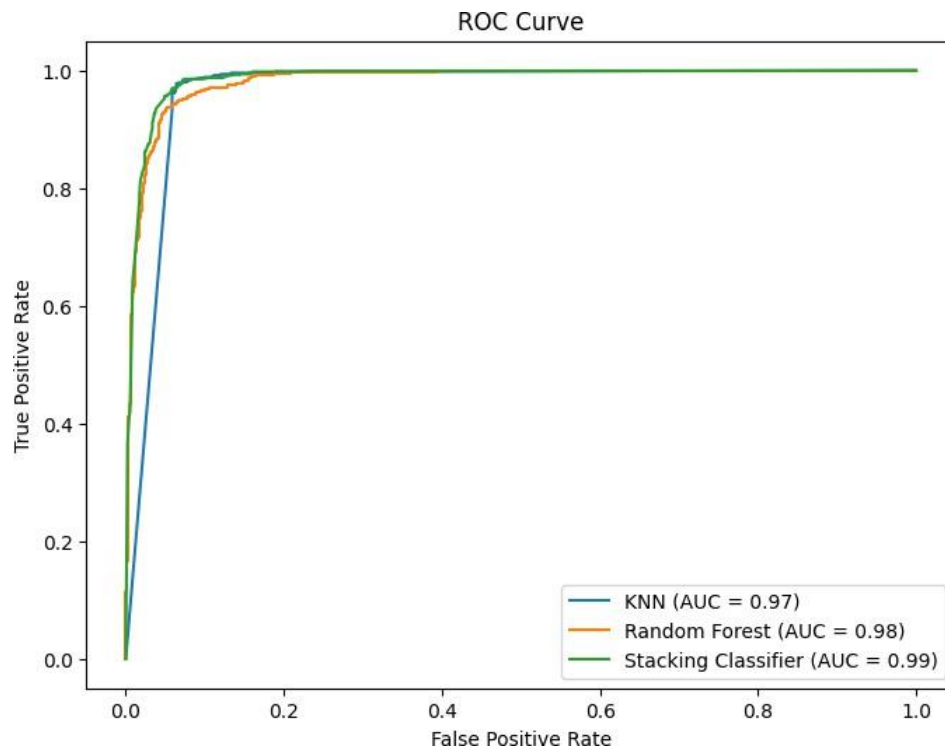


Figure 4.5: ROC Curve

Analysis of ROC Curves

- 1. KNN (Blue Curve, AUC = 0.97):** This model is really good at predicting things, with a very high accuracy in telling the difference between two categories. The line on the graph goes up fast on the left side, which shows that the model is good at identifying true positives and few false positives when the threshold is set low.
- 2. Random Forest (Orange Curve, AUC = 0.98):** Slightly better than KNN, the Random Forest model exhibits a similarly steep ascent in the ROC curve, reaching close to the top-left corner, which is indicative of outstanding discriminative ability. Its AUC of 0.98 means that it nearly perfectly distinguishes between patients with and without heart disease.
- 3. Stacking Classifier (Green Curve, AUC = 0.99):** The Stacking Classifier, integrating outputs from multiple models, shows the best performance among the three. With an AUC of 0.99, it demonstrates almost perfect classification capability. The curve closely hugs the left axis and the top border of the ROC space, reflecting a very high true positive rate alongside a very low false positive rate.

The ROC curves and their respective AUC values confirm that all three models perform exceptionally well in heart disease prediction, but the Stacking Classifier stands out with the highest AUC, suggesting that it is the most reliable model for diagnosing heart disease in this study. High AUC values across all models also indicate that each model manages to achieve a significant separation between the positive and negative cases, a critical requirement in medical diagnostics where the cost of false negatives can be very high.

Using these models, especially the Stacking Classifier, can really help doctors find heart disease early and diagnose it accurately. This can then help them provide the right treatment at the right time. The Stacking Classifier is really good at telling things apart. This might help stop patients from getting treatment they don't actually need, and save money on healthcare. It could also stop patients from worrying about being told they're sick when they're not.

Chapter 5

Conclusion and Recommendation

This study looked at using new computer programs to better predict heart disease. The results have shown that the new model can predict better and last longer than the old method.

5.1 Main Findings and Contribution

Main Findings of the Research:

1. Enhanced Predictive Accuracy: The model that used K-Nearest Neighbors, Random Forest, and Gradient Boosting Classifier together was better than each model on its own. The model created by combining different methods was better at predicting, with 95% accuracy. 27% more than the KNN and RF models by themselves.

2. Improved Metric Scores: Not only did the stacking ensemble model show it was right, but it also got better in other important ways. It was right 95% of the time at finding sick people and not making errors. This study proves that the model needs to be able to make accurate predictions in order to be useful in healthcare.

3. Robustness and Generalizability: using a special tool to make predictions better helps fix mistakes and differences in each model. This method makes the predictability model stronger and more flexible, so it can work well with different types of data

Contribution of the Research:

1. Advancements in Medical Informatics: This research is important for medical world because it offers a very easier method to predict heart disease. The stack ensemble model will help doctors to predict heart diseases earlier, which will help them to make the proper treatment

2. Framework for Future Research: The results are a great way start for future research on using machine learning for medical problems. This study found that using a combination of algorithms can help understand difficult medical information better.

3. Potential for Broader Applications: Stacking methods can be used in many different sectors of medical world. This method can be used to predict different diseases. It can also be used in healthcare to make better predictions, which can help doctors diagnose and treat people more accurately.

5.2 Future Recommendations

Although the current model shows a noticeable increase in predicting accuracy, upcoming studies can investigate various ways to improve the model's performance and usefulness:

- 1. Incorporation of More Diverse Data:** In the future, researchers could study more things like people's age, genes, and how they live to learn more about patients. This could help people understand better the complicated features of heart diseases.
- 2. Exploration of Additional Ensemble Techniques:** Trying different ways of grouping models together, like boosting or bagging, with different base models could help us improve how well our models work
- 3. Application of Deep Learning:** Because medical information can be very complex, it could be useful to study deep learning methods. These methods can find detailed patterns in large sets of data and may be better than traditional machine learning methods.
- 4. Real-World Testing and Validation:** Testing the model in a real medical setting to make sure it works well in a real-life situation is important to do next. This means checking how good the model is over time and if it can handle new information well.
- 5. Ethical and Privacy Considerations:** As models become more integrated into healthcare systems, addressing ethical and privacy concerns related to patient data will be essential. Future research should also focus on ensuring that these models adhere to ethical standards and regulations

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Chapter A

Appendix

A.1 Example of Tables

Table A.1: Data Description of the dataset

Attribute	Data Type	Description
ID	int64	Distinct ID for each patient
Gender	Categorical	Identifies the patient's gender
Age	float64	Age of the patient
Work Type	Categorical	Informs about the working category
Ever married	Categorical	Mention about marital status
Residence Type	Categorical	Indicate the living environment
Heart Disease	int64	Whether or not the patients have cardiac disease
Hypertension	int64	The presence or absence of hypertension in the patients
BMI	float64	Represent the patient's Body mass index (BMI) value
Glucose Level	float64	It indicates the average level of glucose
Stroke	int64	Informs about the prior stroke
Smoking Status	Categorical	Gives the patient's smoking scenario

Table A.2: Sample of the processed stroke prediction dataset:

gender	age	h_tension	h_disease	married	work	Residence	avg_glucose_level	bmi	smoking	stroke
1	67.0	0	1	0	0	0	228.69	36.6	2	1
1	80.0	0	1	0	0	0	105.92	32.5	2	1
0	49.0	0	0	1	0	1	171.23	34.4	3	1
1	79.0	1	0	0	1	1	174.12	24.0	0	1
1	81.0	0	0	0	0	0	186.21	29.0	2	1
0	13.0	0	0	1	2	1	103.08	18.6	1	0
1	81.0	0	0	1	0	0	125.20	40.0	2	1
1	35.0	0	0	1	1	1	82.99	30.6	2	1
1	51.0	0	0	0	0	0	166.29	25.6	2	1
0	44.0	0	0	1	3	0	85.28	26.2	1	0

Table A.3: Algorithm Parameters

Model	Parameters
KNN	n.neighbors=3, weights= distance
Random Forrest	min_samples leaf=4, min_samples split= 5,n_estimators=300
Gradient Boosting Classifier	n_estimators=100, learning rate=0.05, max depth=3

Table A.4: Performance Comparison of Predictive Models

Model	Accuracy	Precision	Recall	F1 Score
KNN	93.78%	93%	94%	93%
Random Forest	93.62%	93%	94%	94%
Stacking Classifier	95.27%	95%	95%	95%

A.2 Example of figures

The Figure represents Methodology Block Diagram:

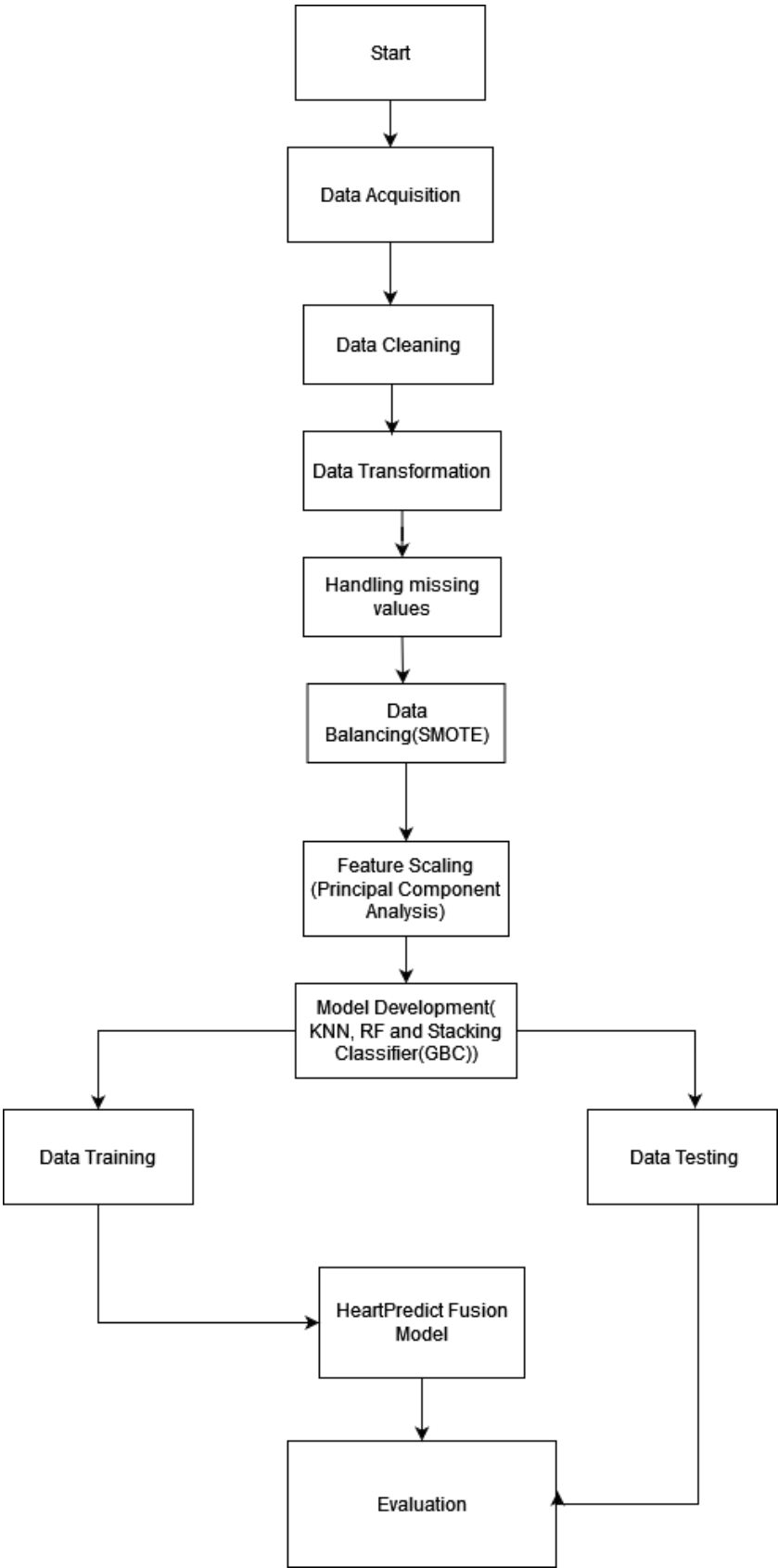


Figure A.1: Methodology Block Diagram

The Figure represents Proposed Model Architecture:

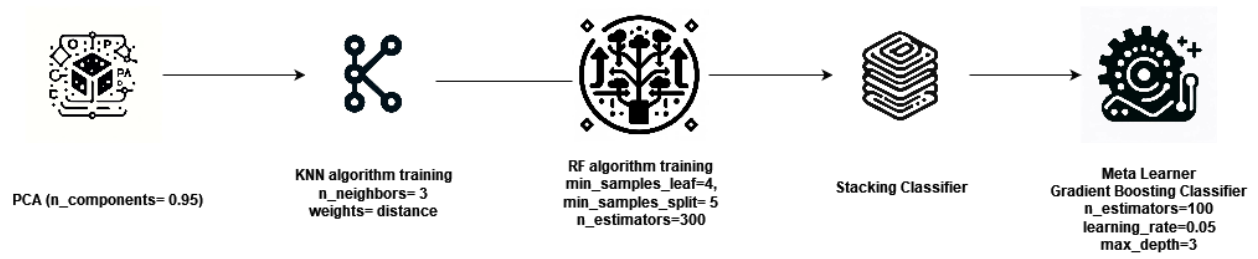


Figure A.2: Proposed Model Architecture

The Figure represents Correlation matrix:

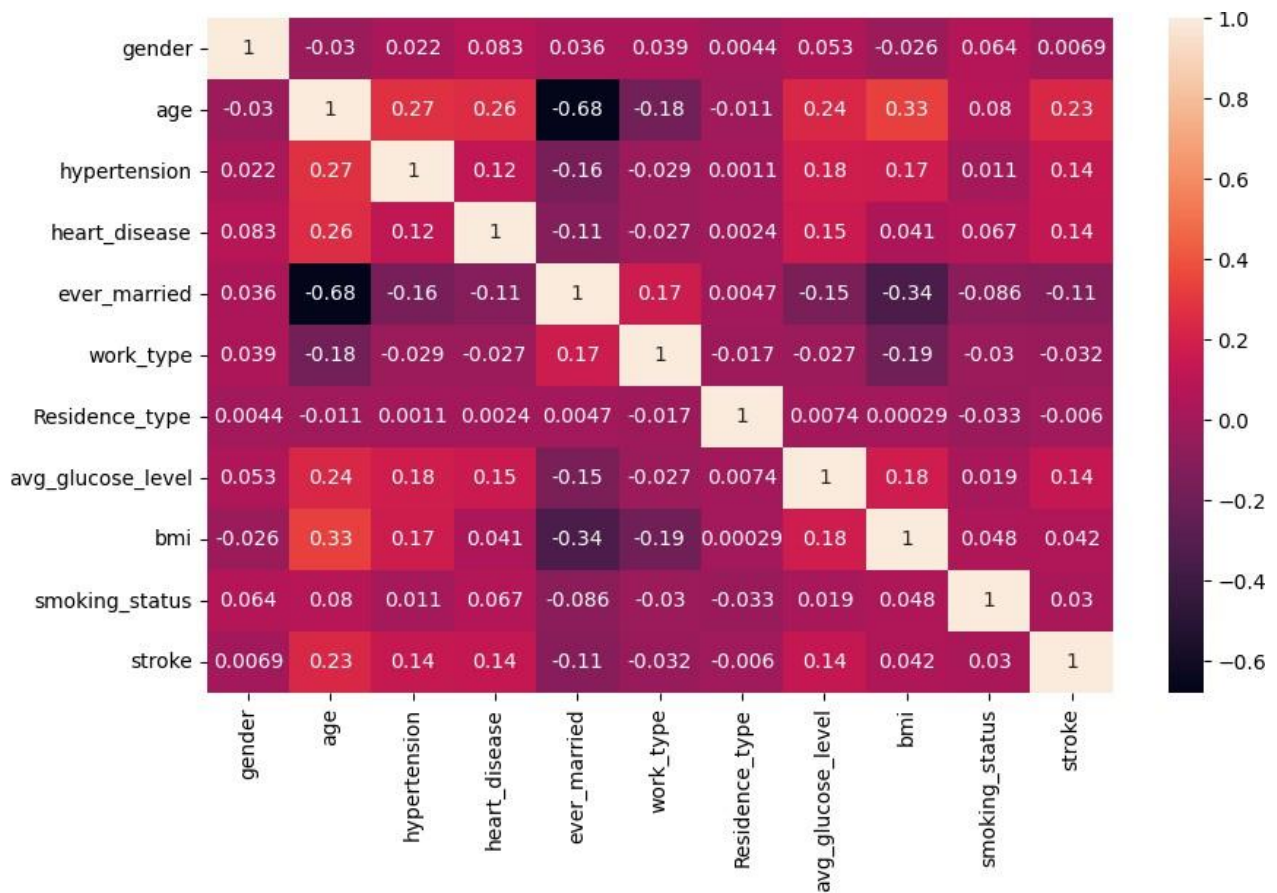


Figure A.3: Correlation matrix

The Figure represents KNN Confuion Matrix:

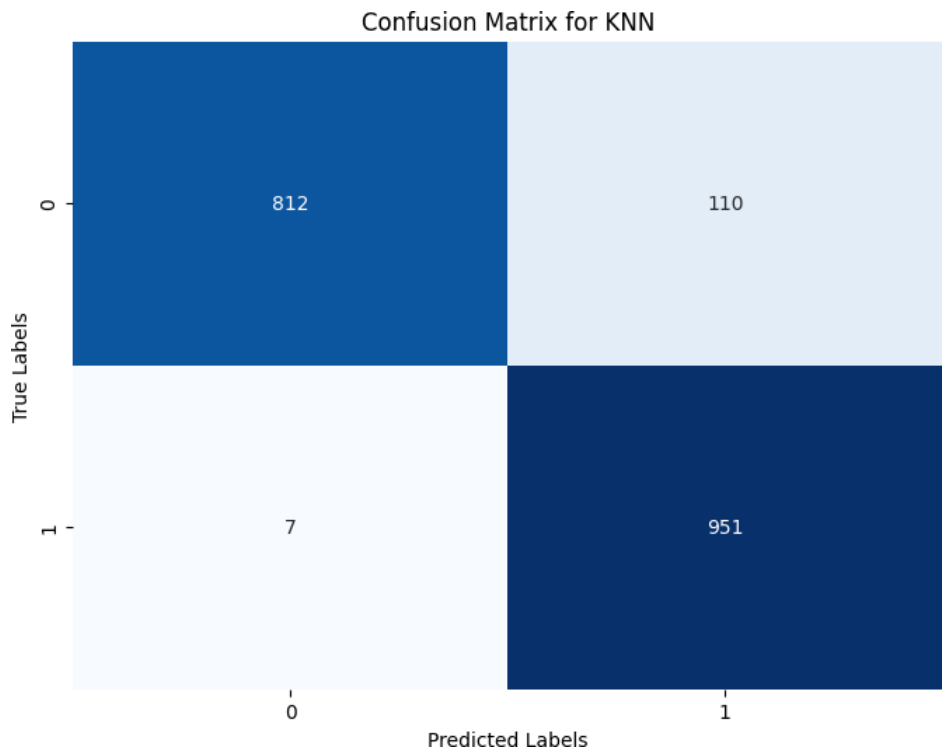


Figure A.4: KNN Confuion Matrix

The Figure represents Random Forrest Confusion Matrix:

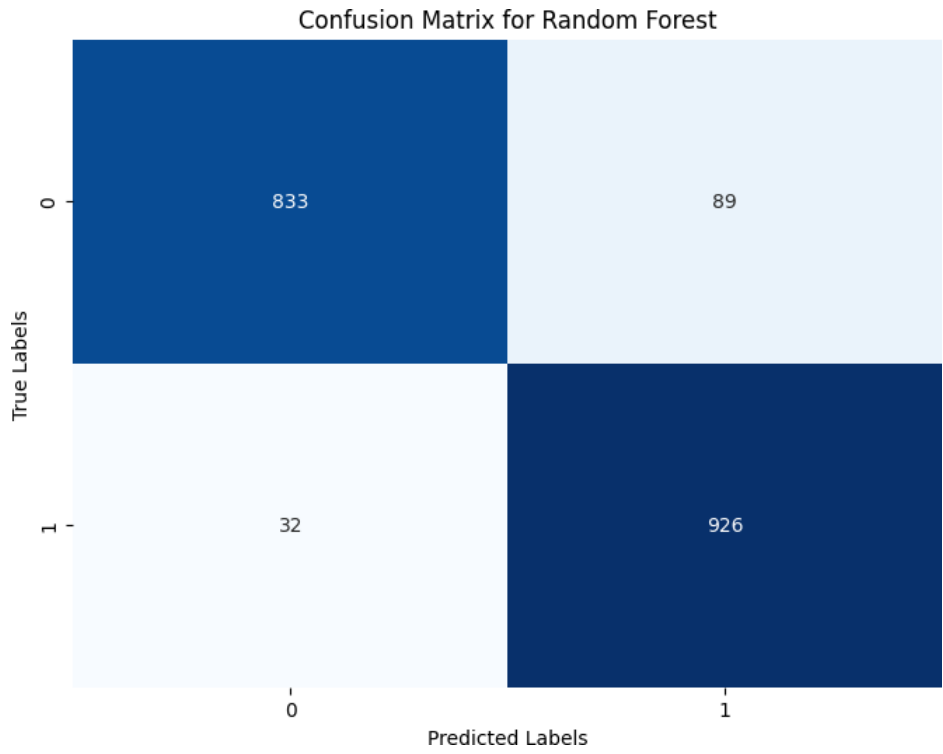


Figure A.5: Random Forrest Confusion Matrix

The Figure represents Stacking Classifier Confusion Matrix:

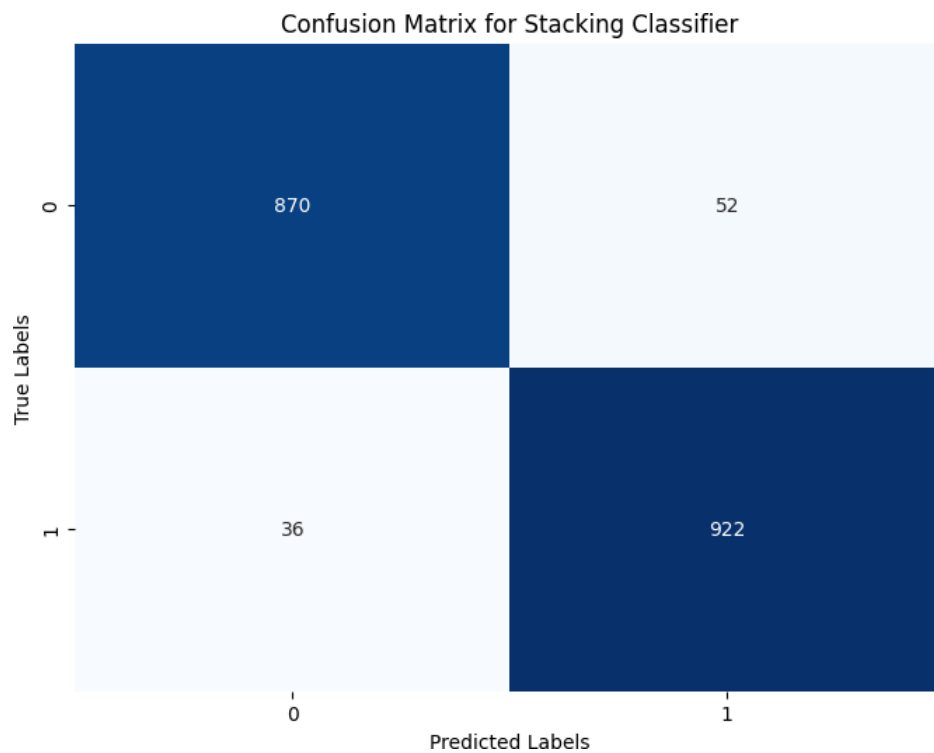


Figure A.6: Stacking Classifier Confusion Matrix

The Figure represents ROC Curve:

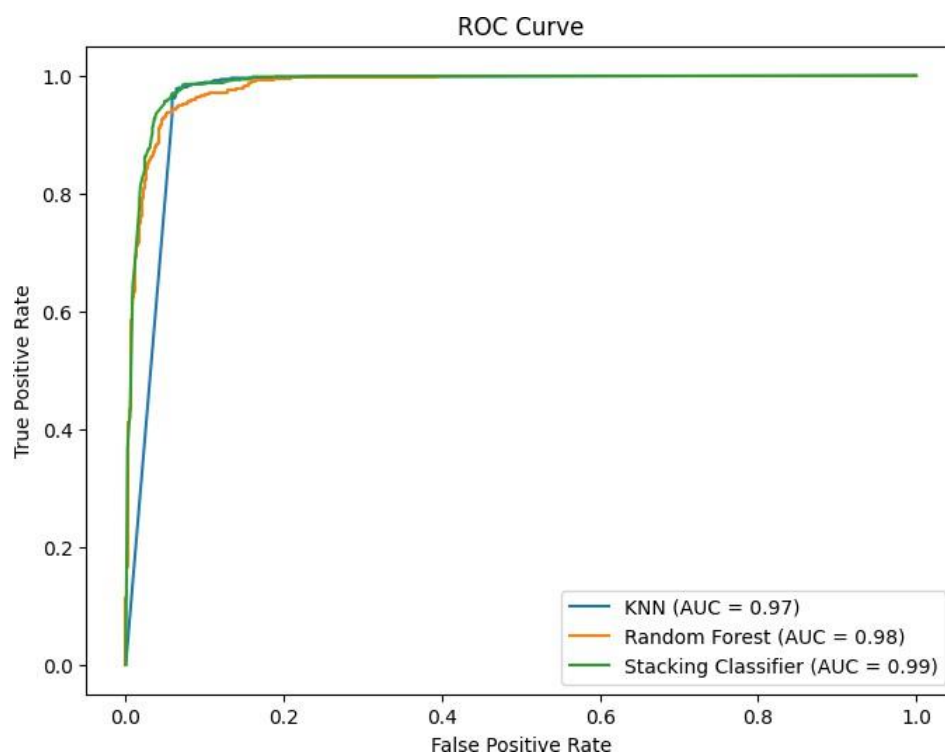


Figure A.7: ROC Curve

Chapter B

Example of Code

1. Importing the libraries:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from imblearn.over_sampling import SMOTE
from collections import Counter
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import GridSearchCV
from sklearn.metrics import accuracy_score, roc_auc_score, f1_score, recall_score, confusion_matrix
from sklearn.model_selection import cross_val_score
from prettytable import PrettyTable
from sklearn.ensemble import GradientBoostingClassifier, StackingClassifier
from sklearn.decomposition import PCA
from sklearn.pipeline import make_pipeline
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report
```

2. Loading the dataset:

```
df = pd.read_csv("healthcaredatasetstrokedata.csv")
```

3. Exploratory data analysis:

```
df.info()
```

```
df
```


3. Removing the first column, "ID," since it doesn't contribute valuable information:

```
df = df.drop('id', axis=1)
```

4. Check for duplicates:

```
duplicates = df[df.duplicated()]  
duplicates
```

5. There were 201 entries with missing values for BMI. eliminating the rows from the DataFrame, df, where BMI is not available:

```
df.dropna(subset=['bmi'], inplace=True)
```

6. Map numerical values to the categorical variables

```
Unique_gender = df['gender'].unique()  
print("Categories are:", Unique_gender)
```

```
category_counts = df['gender'].value_counts()  
category_counts
```

```
df.drop(df[df['gender'] == 'Other'].index, inplace=True)
```

```
gender_mapping={'Female': 0, 'Male': 1}  
df['gender']=df['gender'].map(gender_mapping)
```

```
ever_married_mapping={'Yes': 0, 'No': 1}  
df['ever_married']=df['ever_married'].map(ever_married_mapping)
```

```
work_type_mapping={'Private': 0, 'Selfemployed': 1, 'children': 2, 'Govt job': 3, 'Never worked': 4}  
df['work_type']=df['work_type'].map(work_type_mapping)
```

```
Residence_type_mapping={'Urban': 0, 'Rural': 1}
```

```
df['Residence_type']=df['Residence_type'].map(Residence_type_mapping)
```

```
smoking_status_mapping='never smoked': 0, 'Unknown': 1, 'formerly smoked': 2, 'smokes': 3
df['smoking_status']=df['smoking_status'].map(smoking_status_mapping)
```

7. Feature selection

```
x = df.drop(['ever married', 'stroke'], axis=1)
y = df.stroke
```

8. Standardize the data

```
sc = StandardScaler()
x = sc.fit_transform(x)
```

9. Correlation matrix of variables.

```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(),annot=True)
```

10. Apply SMOTE to balance the class distribution

```
smote = SMOTE(random_state=42)
x_resampled, y_resampled = smote.fit_resample(x, y)
fig, axes = plt.subplots(1, 2, figsize=(10, 5))
```

```
sns.countplot(x=y, ax=axes[0])
axes[0].set_title('Class Distribution Before SMOTE')
axes[0].set_xlabel('Target Variable, stroke')
axes[0].set_ylabel('Count')
```

```
sns.countplot(x=y_resampled, ax=axes[1])
axes[1].set_title('Class Distribution After SMOTE')
axes[1].set_xlabel('Target Variable, stroke')
axes[1].set_ylabel('Count')
plt.tight_layout()
plt.show()
```

11. Split the resampled data into training and testing sets

```
x_train, x_test, y_train, y_test = train_test_split(x_resampled, y_resampled, test_size=0.2, random_state=42)
```

12. Initialize KNN classifier with parameters, Train the KNN model on the training set, Predict on the training set

```
knn_model = make_pipeline(StandardScaler(), PCA(n_components=0.95), KNeighborsClassifier(n_neighbors=3, weights='distance'))
```

```
knn_model.fit(x_train, y_train)
```

```
y_train_pred = knn_model.predict(x_train)
```

13. Evaluate the KNN model on the training set, Predict on the testing set, Evaluate the KNN model on the testing set

```
train_accuracy = accuracy_score(y_train, y_train_pred)
```

```
y_test_pred = knn_model.predict(x_test)
```

```
test_accuracy = accuracy_score(y_test, y_test_pred)
```

```
print("Accuracy of KNN:", accuracy_score(y_test, y_test_pred))
```

14. Initialize RF classifier with parameter, Train the RF model on the training set, Predict on the training set

```
rf_model = make_pipeline(StandardScaler(), PCA(n_components=0.95), RandomForestClassifier(min_samples_leaf=min_samples_split=5, n_estimators=300))
```

```
rf_model.fit(x_train, y_train)
```

```
y_test_pred_rf = rf_model.predict(x_test)
```

15. Evaluate the Random Forest model on the testing set, Print the accuracy

```
test_accuracy = accuracy_score(y_test, y_test_pred_rf)
```

```
print("Accuracy of Random Forrest:", test_accuracy)
```

16. Initialize Base model with parameters, Initialize Final Estimator with parameter, Initialize Stacking Classifier , Training , Prediction, Accuracy Evaluation

```
base_models = [
('knn', make_pipeline(StandardScaler(), PCA(n_components=0.95), KNeighborsClassifier(n_neighbors=3,
p=1, weights='distance'))),
('rf', make_pipeline(StandardScaler(), PCA(n_components=0.95), RandomForestClassifier(min_samples_leaf=4,
min_samples_split= 5, n_estimators=300 ))) ]
```

```
final_estimator = (GradientBoostingClassifier(n_estimators=100, learning_rate=0.05, max_depth=3))
```

```
stack = StackingClassifier(estimators=base_models, final_estimator=final_estimator, cv=5)
```

```
stack.fit(x_train, y_train)
```

```
stack_pred = stack.predict(x_test)
```

```
stack_accuracy = accuracy_score(y_test, stack_pred)
```

```
print("Accuracy score using Stacking:", stack_accuracy)
```

17. Define a function to plot confusion matrix

```
def plot_confusion_matrix(cm, title):
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', cbar=False)
plt.title(title)
plt.xlabel('Predicted Labels')
plt.ylabel('True Labels')
plt.show()
```

18. Compute confusion matrix for KNN

```
knn_conf_matrix = confusion_matrix(y_test, y_test_pred)
plot_confusion_matrix(knn_conf_matrix, 'Confusion Matrix for KNN')
```

19. Compute confusion matrix for Random Forest

```
rf_conf_matrix = confusion_matrix(y_test, y_test_pred_rf)
plot_confusion_matrix(rf_conf_matrix, 'Confusion Matrix for Random Forest')
```

20. Compute confusion matrix for Stacking Classifier

```
stack_conf_matrix = confusion_matrix(y_test, stack_pred)
plot_confusion_matrix(stack_conf_matrix, 'Confusion Matrix for Stacking Classifier')
```

21. Define a function to remove the support column from the classification report `def re-`

```
move_support_column(report):
```

```
lines = report.split("\n")
```

```
new_lines = []
```

```
for line in lines:
```

```
if line.strip().startswith('support'):
```

```
continue
```

```
new_lines.append(line)
```

```
return "\n".join(new_lines)
```

22. Generate classification report for KNN

```
knn_report = classification_report(y_test, y_test_pred, zero_division=0)
```

```
knn_report = remove_support_column(knn_report)
```

```
print("Classification Report for KNN:")
```

```
print(knn_report)
```

23. Generate classification report for Random Forest

```
rf_report = classification_report(y_test, y_test_pred_rf, zero_division=0)
```

```
rf_report = remove_support_column(rf_report)
```

```
print("Report for Random Forest:")
```

```
print(rf_report)
```

24. Generate classification report for Stacking Classifier `stack_report = classification_report(y_test,`

```
stack_pred, zero_division=0)
```

```
stack_report = remove_support_column(stack_report)
```

```
print("Report for Stacking Classifier:")
```

```
print(stack_report)
```